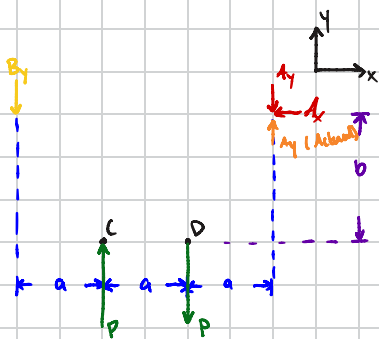




$$P = 25 \text{ kN} \quad a = 0.4 \text{ m} \quad b = 0.3 \text{ m} \quad B_y = ? \quad A_y = ?$$

$$\sum M_A = 0 \quad A_x = 0$$

$$\sum M_A = B_y(1.2) - P(0.8) + P(0.4) = 0$$

$$\sum M_A = B_y(1.2) - 25(0.8) + 25(0.4) = 0$$

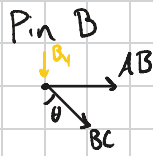
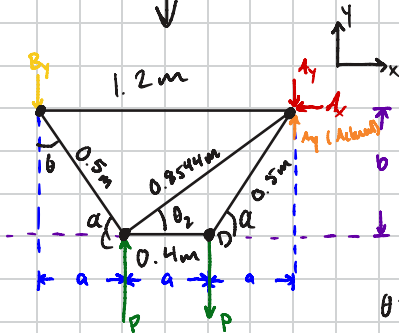
$$B_y = 8.33 \text{ kN}$$

$$\sum F_y = 0$$

$$\sum F_y = A_y - B_y + P = 0$$

$$\sum F_y = A_y - 8.33$$

$$A_y = -8.33 \text{ kN}$$



$$\sum F_y = -B_y - BC \cos(\theta) = 0$$

$$\sum F_y = -8.33 - BC \cos(53.13) = 0$$

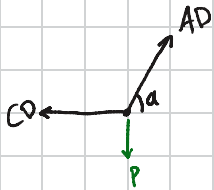
$$BC = -13.89 \text{ kN Compression}$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right) = 53.13^\circ$$

$$\sum F_x = AB - 13.89 \sin(53.13) = 0$$

$$AB = 11.11 \text{ kN Tension}$$

Pin D $\alpha = 36.87^\circ$

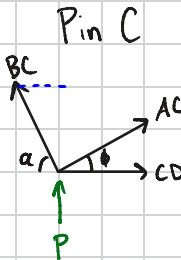


$$\sum F_y = -25 + AD \sin(36.87)$$

$$AD = 41.67 \text{ kN Tension}$$

$$\sum F_x = -CD + 41.67 \cos(36.87)$$

$$CD = 33.33 \text{ Tension}$$



$$\sin(36.87) = \frac{0.3}{0.5} = 0.6$$

$$\cos(36.87) = \frac{0.4}{0.5} = 0.8$$

$$\theta_2 = \tan^{-1}\left(\frac{0.3}{0.4}\right) = 20.56^\circ$$

$$\sin(20.56) = \frac{0.3}{0.8544} = 0.3511$$

$$\cos(20.56) = \frac{0.8}{0.8544} = 0.9363$$

Truss Cross-section

$$F_{\max} = 47.46 \text{ kN}$$

$$S_y = 170 \text{ ksi} = 170,000 \text{ psi} \times \frac{6894.76 \text{ Pa}}{\text{psi}} = 1172 \text{ MPa}$$

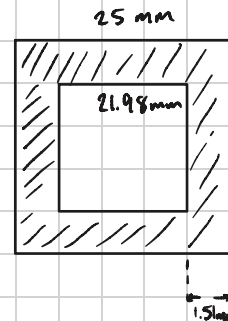
$$SF = 3.5$$

$$A_{\text{req}} = \frac{47.46 \text{ kN} \cdot 3.5}{1172 \text{ MPa}} = 0.0001417 \text{ m}^2 \times \left(\frac{1000 \text{ mm}}{1 \text{ m}}\right)^2 = 141.7 \text{ mm}^2$$

$$\sum F_y = (-13.89) \sin(36.87) + AC \sin(20.56) + 25 = 0$$

$$AC = \frac{13.89 \sin(36.87) - 25}{\sin(20.56)}$$

$$AC = -47.46 \text{ kN compression}$$



Truss weight

$$\text{Density} = 0.278 \text{ lb/in}^3$$

$$\text{Total Member Length} = 3.4544 \text{ m} = 3454.4 \text{ mm}$$

$$\text{Volume} = 141.7 \text{ mm}^2 \cdot 3454.4 \text{ mm} = 489488.48 \text{ mm}^3 \times \left(\frac{1 \text{ in}}{25.4 \text{ mm}}\right)^3 = 29.87 \text{ in}^3$$

$$\text{Total weight} = 8.304 \text{ lbs}$$

Pin Shear

$$\tau_{\text{yield}} = 0.5 S_y$$

Tensile yield strength: 170 ksi

Safety Factor: 4

Single Shear

$$A = \frac{\pi d^2}{4}$$

$$4A = \pi d^2$$

$$d = \sqrt{\frac{4A}{\pi}}$$

$$d = \sqrt{\frac{4(324)}{\pi}}$$

$$d = 20.31 \text{ mm}$$

$$\tau = \frac{0.5 S_y}{SF}$$

$$A_{\text{pin}} = \frac{V}{\tau_{\text{all}}} = \frac{V \cdot SF}{0.5 S_y}$$

$$A_{\text{pin}} = \frac{47470 \text{ N} \cdot 4}{0.5(117210 \text{ N/mm}^2)} = 0.000324 \text{ m}^2 = 324.3 \text{ mm}^2$$

